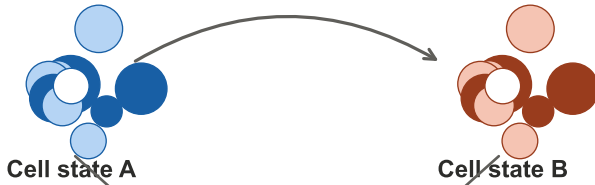


CKI: A Cell-state Kinetic Index

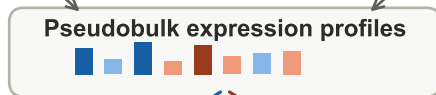
Quantifying selective transcriptomic remodeling from expression distributions

compare expression profiles

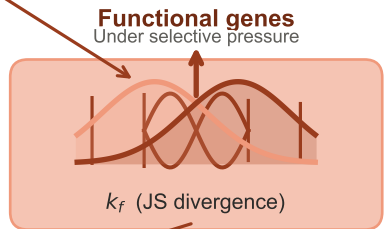
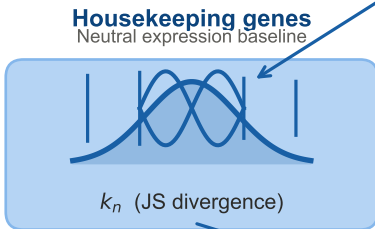
1



2



3



4

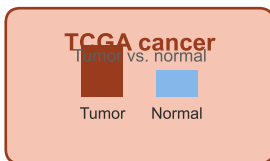
$$\omega = k_f/k_n$$

$\omega > 1 \rightarrow$ selection exceeds drift

5



Applications



Key insight:

ω distinguishes functional remodeling from neutral transcriptomic drift